“Are you feeling sick?” – A systematic literature review of cybersickness in virtual reality

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Cybersickness (CS), also known as visually induced motion sickness (VIMS), is a condition that can affect individuals when they interact with virtual reality (VR) technology. This condition is characterized by symptoms such as nausea, dizziness, headaches, eye fatigue, and so on, and can be caused by a variety of factors. Finding a feasible solution to reduce the impact of CS is extremely important as it will greatly enhance the overall user experience and make VR more appealing to a wider range of people. We have carefully compiled a list of 223 highly pertinent studies to review the current state of research on the most essential aspects of CS. We have provided a novel taxonomy that encapsulates various aspects of CS measurement techniques found in the literature. We have proposed a set of CS mitigation guidelines for both developers and users. We have also discussed various CS-inducing factors and provided a taxonomy that tries to capture the same. Overall, our work provides a comprehensive overview of the current state of research in CS with a particular emphasis on different measurement techniques and CS mitigation strategies, identifies research gaps in the literature, and provides recommendations for future research in the field.

CCS Concepts: • **Human-centered computing → VR;**

Additional Key Words and Phrases: Virtual reality, cybersickness, systematic review, cause, measurement, mitigation, guidelies

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1 Introduction

Virtual reality (**VR**) technology has been rapidly advancing in recent years, with many new applications and devices becoming available to consumers. There are significant initiatives to incorporate VR technology into numerous industries, e.g., gaming, film, education, and the medical sector. The importance of VR is further underscored in settings when social interaction is constrained, such as those brought about by the most recent COVID-19 outbreak. The recent technology trends suggest that the world is heading toward immersive technologies. Seeing the potential for rapid growth in this market, many big technological giants, like Meta, Microsoft, and Google, have started investing heavily in immersive technologies. However, some VR users may experience uncomfortable symptoms resembling those of **motion sickness** (**MS**). This phenomenon in the virtual environment has been termed ***Cybersickness*** (**CS**) [161]. A few of the symptoms that characterize CS are as follows—nausea, headache, excessive sweating, dryness of mouth, fullness of stomach, vertigo, ataxia, vomiting, dizziness, disorientation, eye strain, fatigue, strange hand-eye coordination, and so on [122, 124, 227]. These undesired symptoms can be caused by a variety of factors, including the mismatch between the visual and vestibular systems, the latency of the device, and the design of the VR experience. Sherman [211] suggested that a person may get MS if they consistently experience sensory input that does not match their expectations.

The term *Cybersickness* was first coined in the early 1990s, around the same time that VR technology was first becoming widely available. Early research on CS focused on identifying the symptoms and causes of the condition, as well as exploring potential ways to mitigate or prevent it completely. The resemblance between the symptoms of MS and CS laid the groundwork for initial theories exploring the origins of CS. Despite this superficial similarity, it becomes clear upon closer examination that CS and MS exhibit distinct symptom profiles. Recognizing these differences, Stanney et al. [222] suggested that MS and CS should be considered separate entities in terms of their causative factors, symptomatology, and potentially their mitigation strategies. This distinction emphasizes the need for tailored approaches to understanding and addressing each condition within the context of VR and traditional motion-induced discomfort. Researchers have employed various strategies to measure CS. The measurement techniques of CS have evolved from simple questionnaires to more complex and sophisticated techniques, reflecting the growing understanding of the condition and the need for better methods to assess it. Researchers have been continuously trying to come up with novel approaches to mitigate the effects of CS to create a better user experience and thus increase the adoption rate of VR technology.

There are several review articles on CS in VR. One such review was conducted by Davis et al. [39], which employed a systematic approach to examine CS with a specific emphasis on its measurement. Notably, this review was published in 2014, and since its publication, considerable advancements have occurred in the field of VR measurement. Researchers have developed more sophisticated measurement tools and methodologies to quantify the subjective experiences and physiological responses associated with VR-induced discomfort. These advancements include the refinement of self-report questionnaires, the **Virtual Reality Symptom Questionnaire** (**VRSQ**), to better capture the nuances of CS symptoms. Additionally, improvements in the integration of physiological sensors, such as **heart rate variability** (**HRV**) monitors and eye-tracking devices, have enabled more objective and comprehensive assessment of users' physiological reactions to VR stimuli. Moreover, advancements in the field of **Machine Learning** (**ML**) have facilitated the interpretation and validation of measurement outcomes, contributing to a deeper understanding of the mechanisms underlying CS and other VR-related phenomena. Several measurement techniques and mitigation strategies have been proposed in the literature, warranting an updated analysis of the field. Furthermore, Rebenitsch and Owen [195] conducted a review that explored

diverse factors that can contribute to CS, such as the **field of view** (**FOV**), navigation, and visual display type, while stressing the necessity for further research to achieve a more comprehensive understanding of this condition. To remain current with the latest advancements in the field of CS, literature reviews have also been conducted by Chang et al. [28], Caserman et al. [26], and Saredakis et al. [205]. However, the focus of these reviews primarily centered around the causes of CS, with limited coverage of measurement and mitigation techniques of CS. It is important to note that several other reviews in the field of CS are available, each with a particular emphasis on a specific aspect of measurements in CS. For instance, Yildirim and Caglar [266] conducted a review of various **deep learning** (**DL**) based approaches for the classification of CS using **electroencephalogram** (**EEG**) signals, while Yang et al. [263] reviewed various ML based approaches used to detect/predict CS. Furthermore, Li et al. [130] reviewed the electrostimulation-based methods for CS mitigation. Some reviews were also conducted from the perspective of gender [154], latency [225], and user susceptibility to CS [231]. However, these reviews fail to provide a holistic view of all the important aspects of CS, namely its causes, measurement, and mitigation strategies.

To address these gaps in the literature, our systematic literature review provides a comprehensive analysis of all three aspects of CS, namely its causes, measurement, and mitigation. Our survey is motivated by the realization that, despite significant strides in understanding CS, there remains a fragmented landscape of research that often focuses on isolated aspects of the phenomenon. Previous reviews have offered valuable insights into the causes, measurements, and some mitigation strategies of CS, yet there's a noticeable lack of a unified view that encompasses the full spectrum of CS research. In particular, the rapid evolution of VR technologies and their applications has outpaced the current understanding of CS, necessitating a fresh and holistic examination. The following research questions were formulated for the current study:

- RQ1: (a) What are the leading causes of CS in VR environments? (b) How do they impact users across different applications?
- RQ2: How can CS be accurately measured and assessed, taking into account the evolving nature of VR technologies and user interaction paradigms?
- RQ3: (a) What strategies and interventions have proven effective in mitigating the effects of CS? (b) How can the measurements techniques be applied across the diverse spectrum of VR applications?
- RQ4: (a) What are the emerging research gaps in the study of CS? (b) How can the gaps inform future inquiries and technological developments?

The main contributions of this review are summarised below.

- A carefully curated list of 223 research papers was used to critically analyze various aspects of CS, namely causes, measurement, and mitigation of CS. Analyzing this huge collection of research papers allowed us to gather critical insights about various aspects of CS that are often not captured by other similar studies.
- We have proposed a novel taxonomy that encompasses several facets of CS measurement techniques. This will act as an overview of several measuring methodologies (as of February 2023) and assist researchers in making informed decisions with regard to the measurement of CS.
- We have proposed a mitigation guideline for CS. Based on the available literature and our critical thinking, we have provided a set of guidelines for both the developers and the end users. We have also discussed various mitigation strategies found in the literature both from the developer's perspectives and end users' perspectives.
- We have created a comprehensive taxonomy that categorizes these various CS-inducing factors in order to organize better and comprehend them.

— We have identified a number of research gaps in various aspects of CS research during the course of our systematic literature review and have also provided some proposals for further investigation.

The remainder of this article is organized in the following manner: Section 2 gives a thorough explanation of the review process that was followed to conduct this systematic literature review. Section 3 is a discussion on various CS-inducing factors. Section 4 discusses various measurement techniques of CS, followed by a discussion on various strategies used to mitigate CS both from developers' as well as users' perspectives (Section 5). Section 6 discusses various research gaps present in the literature. This section also contains our proposed set of mitigation guidelines for both users as well as developers. Section 7 concludes our article.

2 Systematic Review Methodology

Transitioning from the broad overview provided in the introduction, this section outlines the methodology employed to ensure a robust and thorough review of the existing literature on CS. We aim to provide an extensive review of the various causes, measurement techniques, and mitigation strategies of CS. To ensure a comprehensive literature review, we utilized the following well-established databases with a history of publishing VR research—ACM DL, IEEE, PubMed, and Web of Science. Through this approach, we aim to analyze the existing body of research on this phenomenon and to identify gaps in knowledge that warrant further investigation. By synthesizing and critically evaluating the available evidence, this review can contribute to a better understanding of CS and inform the development of effective prevention and mitigation strategies. Table 2 displays the search strings that were utilized in the database search. We reviewed the papers in the last decade (from 2014 till February of 2023) as a huge boom in VR research was observed in the last decade. We found a total of 1005 records with the search strings, which were subsequently subjected to the first stage of screening following the PRISMA systematic review protocol [136, 175]. We also extracted 80 more papers from the Google Scholar search. Out of 1085 records, 180 were found to be duplicates. We excluded 22 poster/workshop papers that were less than 3 pages (double column) in length. We also excluded 4 papers that were not available in public or not in the English language. The flow diagram of the PRISMA method is shown in the Figure 1. To determine eligibility, a panel of 3 raters independently assessed the titles and abstracts of the records. Prior to proceeding, a calibration exercise was conducted, whereby the raters rated 150 papers to ensure consistency and reliability. Agreement among the raters was found to be 97.3%. During this initial screening phase, publications were labeled according to the following criteria:

— Relevant, wherein the search terms referred on the title, abstract, and/or conclusion are relevant (providing cause, measurement, or mitigation for CS in VR).
— Not Relevant, wherein the search terms referred on the title, abstract, and/or conclusion are not relevant (Does not provide any information toward causes, measurement, or mitigation for CS in VR).

Out of 879 remaining records, we found 223 relevant articles. We categorized the relevant papers into causes, measurement, and mitigation of CS. We found 21 articles that belonged to both cause and measurement or measurement and mitigation or all three of them. Lastly, forward and backward citation searching was used to incorporate highly relevant and/or cited publications that were not captured by our search [37]. The inclusion and exclusion criteria for the review process are presented in Table 1.

3 Causes of CS: A Complex Interaction of Various Factors

Building on the methodological foundations, we now turn to explore the well-established theories and various factors that contribute to CS, providing a critical understanding of its causes. CS is a

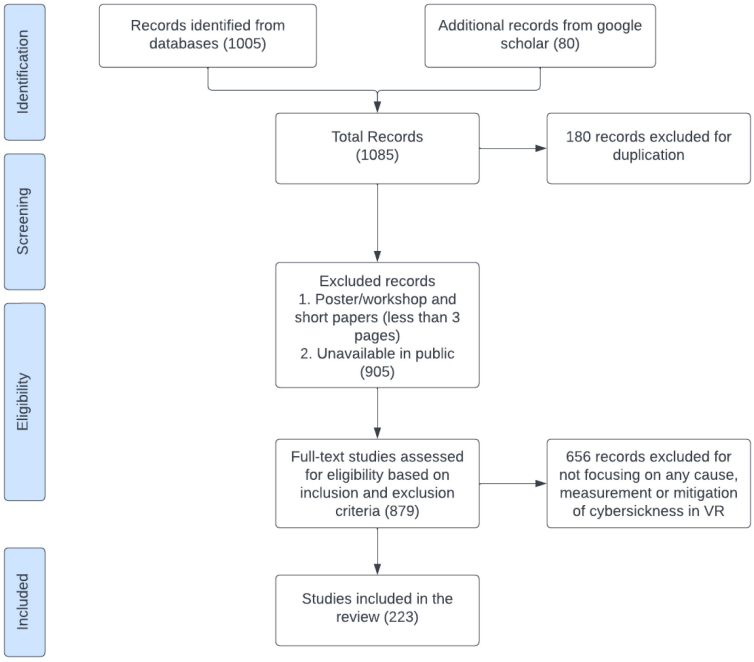

Fig. 1. Prisma flow chart.

Table 1. Inclusion and Exclusion Criteria for Systematic Review on CS

Inclusion Criteria	**Exclusion Criteria**
Peer-reviewed articles published in English.	Studies published in languages other than English.
Research focusing on VR environments and their effects on users, specifically related to CS.	Studies focusing on VR applications without relevance to CS.
Articles that clearly define and measure CS using empirical data.	Conceptual papers or opinion pieces without empirical data.
Studies that test the CS mitigation strategy.	Papers that do not validate specific CS mitigation strategies.
Full paper articles	Short paper or workshops.
Research published between Jan 2014 and Feb 2023, to ensure relevance and timeliness.	Studies published before 2014, to exclude outdated methodologies or findings.
Papers that include a diverse participant demographic (age, gender, VR experience level, etc.) in their studies.	Studies with a narrow participant demographic that limits the generalizability of the findings.
Articles that employ a variety of measurement techniques for CS (subjective questionnaires, physiological measures, etc.).	Studies relying solely on anecdotal evidence or non-standardized measures of CS.

complex phenomenon that is yet to be fully understood. Several theories have attempted to explain its causes, including the sensory conflict theory [14, 17, 173, 193, 194], postural instability theory [200], and poison (also known as evolutionary theory) [235]. Several studies suggested that vection might be responsible for the occurrence of CS [16, 147, 153], although contradictory study exists

Table 2. Databases and Search Strings Used

Database	Search String	Total results
ACM digital library	[[Abstract: "VR"] OR [Abstract: or "Virtual Reality"]] AND [Abstract: "*sickness"] AND [E-Publication Date: (01/01/2014 TO 28/02/2023)]	110
IEEE Xplore Digital Library	("Abstract":"VR" OR "Abstract":"Virtual Reality") AND ("Abstract":"*sickness") Filters Applied: 01/01/2014 - 28/02/2023	295
PubMed Central (PMC)	(("Virtual Reality" [Title/Abstract] OR "VR" [Title/Abstract]) AND ("*sickness" [Title/Abstract])) AND (("2014/01/01"[Date - Publication] : "2023/02/28"[Date - Publication]))	251
Web Of Science	(AB=("Virtual Reality" OR "VR")) AND AB=("sickness") Timespan: 2014-01-01 to 2023-02-28 (Index Date)	349

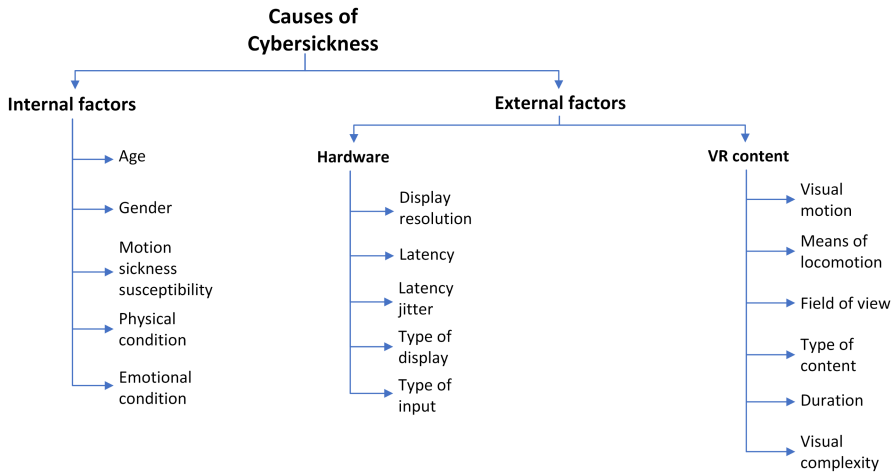

Fig. 2. Taxonomy of causes of CS.

that showed that it is possible to get CS without eliciting vection [88]. Prothero et al.'s rest frame hypothesis [189] suggests that MS is caused by conflicting stationary frames of reference rather than conflicting orientation and motion cues. These theories suggest that conflicting cues from different senses, an unstable posture, and an evolutionary response to perceived poisoning can all contribute to CS. However, none of these theories can fully explain the subjectivity involved in CS or predict its severity in different virtual environments. Our research has identified several key factors that can contribute to the onset of CS. To better organize and understand these factors, we have developed a comprehensive taxonomy that breaks them down into specific categories (As shown in the Figure 2). By examining each of these categories in detail, we hope to shed some light on the complex array of factors that can contribute to CS and provide a more nuanced understanding of this phenomenon.

3.1 Internal Factors

When we refer to "internal factors" in the context of CS, we are specifically talking about a set of individual characteristics and conditions that can contribute to an increased likelihood of experiencing CS. These factors are inherent to the user and can include a variety of different elements.

3.1.1 Age. Studies indicate a nuanced relationship between age and CS susceptibility. For example, while children between 2 and 12 years may exhibit increased sensitivity, this tends to decrease

into young adulthood [248]. A study by Garrido et al. [60] have found that older participants experienced higher levels of CS than younger participants, indicating that age plays a role in the experience of immersive VR. Here, the authors speculated that age-related changes in the vestibular system, which helps to regulate balance and spatial orientation, may play a role. It is noteworthy that CS is uncommon among young children [239]. One reason for this may be that during this developmental stage, young children are frequently carried and moved around. However, a study by [171] found that CS is independent of age when viewing general VR content, but age may be a factor in the experienced level of CS for controllable content. These findings suggest that age plays a critical role in the likelihood of experiencing CS and that developers of VR experiences should be mindful of the specific age-related risks and factors that may contribute to this phenomenon.

3.1.2 Gender. The impact of gender in inducing CS has been the subject of numerous studies, with results varying across the board. While studies by Stanney et al. [221] have found that women are more likely to have more severe cases of CS, other studies revealed that there aren't any discernible differences between male and female participants in terms of the presence and severity of CS. Furthermore, one study [182] noted that the proportion of female participants in VR studies using **head-mounted displays** (**HMDs**) was systematically associated with an increase in simulator sickness, which underlines the importance of sampling representativeness in VR research. There may be a difference in the intensity of CS between male and female participants due to the **interpupillary distance** (**IPD**) of the user, with a greater proportion of females having trouble finding a good fit [220]. This suggests the need for VR displays to have a wider IPD adjustable range, especially among females. Overall, it is apparent that the impact of gender on CS remains a complex issue, and further research is necessary to fully understand this relationship.

3.1.3 Motion Sickness Susceptibility. Research has explored the connection between CS and MS susceptibility. MS occurs when the brain receives conflicting signals from various sensory organs, causing symptoms like nausea and dizziness. As VR technology can produce realistic movements and visuals, users may experience similar sensations to real-world motion, increasing their susceptibility to CS. Several studies have indicated a positive correlation between MS susceptibility and the risk of CS [58, 160, 171, 236]. For example, Gavgani et al. [160] discovered that individuals with higher MS susceptibility experienced more severe CS symptoms when exposed to VR environments. Katsigiannis et al. [90] also found that participants with greater MS symptoms were more prone to CS while using a VR exercise bike. These studies imply that those who are more susceptible to MS are at a greater risk of experiencing CS during VR experiences. Another factor that can influence the likelihood of experiencing CS is prior experience with VR technology. Studies have found that individuals with prior experience with VR technology are less likely to experience CS compared to those without prior experience [60, 270]. For example, a study by Garrido et al. [60] found that individuals who had more exposure to VR technology had lower levels of CS symptoms compared to those with less exposure. Additionally, repeated exposure to VR has been shown to be effective in reducing CS symptoms over time [202].

3.1.4 Physical Conditions. One of the physical conditions responsible for the occurrence of CS is body awareness, which is the ability to attend to and perceive bodily sensations. According to the study by Mittelstädt et al. [162], body awareness is associated with increased susceptibility to CS. Altena et al. [6] showed that poor sleep quality may contribute to the induction of CS. Their research indicated that addressing sleep issues could be important in mitigating CS. Additionally, the study by Kim and Kim [110] suggested that people with greater retinal eccentricity may be more susceptible to CS. Moreover, individuals with neurological conditions, such as multiple

sclerosis, may experience more severe CS symptoms compared to healthy individuals [10]. Besides specific physical conditions, the physical position of the user can also impact susceptibility to CS, according to a study by Xu et al. [262]. They found that participants in the standing posture experienced more CS symptoms compared to those in the seated posture. These findings suggest that considering the physical condition and posture of users may be important in mitigating CS during VR experiences.

3.1.5 Emotional Conditions. Some studies have explored the relationship between CS and the emotional state of the user. Studies [91, 187, 248] have found that anxiety can contribute to the experience of CS in immersive VR environments. One study [238] found that negative emotions, especially anxiety, and frustration, were associated with an increased likelihood of experiencing VIMS. The study suggested that regulating emotions could be a useful strategy for reducing CS. Additionally, it has been found that individual susceptibility to CS is also associated with negative emotional responses to VR experiences, and personality traits such as neuroticism increased the likelihood of experiencing CS [254]. A study by Wang et al. [248] used fuzzy logic to predict CS during VR navigation and found that individual differences in emotional responses, particularly anxiety, and stress, were crucial factors in predicting CS. These studies suggest that understanding individual differences in emotional regulation may be important for reducing the likelihood of CS during VR experiences.

3.2 External Factors

3.2.1 Hardware. The quality of the VR experience of the user is significantly affected by the hardware used. Previous studies initially associated CS symptoms with inferior equipment, assuming that the advancement of technology would alleviate user discomfort [197]. Despite numerous hardware enhancements, CS persists, as some aspects of the device are believed to be contributing to the occurrence of VIMS.

Display resolution: The term "Display resolution" is used to describe the number of pixels that are visible on a screen or monitor. Typically, the display resolution is expressed in terms of how many pixels are present in each row and column of the screen. Higher display resolution means that more pixels are used to display an image, resulting in a clearer and more detailed image. In VR, the display resolution is a critical factor in creating a realistic and immersive experience for the user, as a high resolution can help to reduce the likelihood of CS and other symptoms by providing a more seamless and realistic environment. A study by Wang et al. [246] found that higher render resolution led to a better gameplay experience and reduced CS up to a certain point, with no significant effect on performance. They evaluated CS at resolutions of 1K, 2K, 3K, and 4K and discovered that there was a substantial difference in induced CS between resolutions of 1K and 2K. However, no significant sickness was found among 2K, 3K, and 4K resolutions.

Latency: Latency, or the time delay between a user's movement and the corresponding change in the virtual environment, has been identified as a potential cause of CS. Several studies have investigated the relationship between CS and latency in HMD-based VR. One such study by Palmisano et al. [176] found that reducing latency can help to reduce CS, as it minimizes the discrepancy between the visual and vestibular information. The study also found that participants who had greater differences in head orientation between the virtual and physical environments experienced more CS, due to the conflict between the visual and vestibular systems. Some other studies [20, 93, 234] have confirmed that simulator sickness increased with higher latency, and participants reported more severe symptoms of nausea, disorientation, and eye strain. Additionally, a study by Kuwamura et al. [93] found that higher latency resulted in decreased stability and increased sway-

ing in the forward-backwards direction. These findings suggest that latency is a crucial factor to consider when designing VR systems and experiences, as it can significantly impact CS and overall user experience.

Latency Jitter: Jitter refers to the variability in latency, causing irregularities in the timing of the delivery of data. The delay in response can lead to inconsistencies in the experience, causing discomfort and disorientation for the user. A study by Stauffert et al. [224] showed that latency jitter could significantly affect the incidence of CS. The study found that participants who experienced higher levels of latency jitter were more likely to experience various symptoms of CS. Therefore, it is important to minimize latency jitter in VR systems to improve user experience and reduce the occurrence of simulator sickness.

Type of display: The severity of symptoms due to CS that users may experience can be significantly influenced by the type of display that is utilized in VR systems. A study by Mittelstaedt et al. [163] examined the effects of display type and motion control on CS in a virtual bike simulator. According to the study, those who used an HMD felt more cybersick than those who used a conventional monitor. Another study by Choy et al. [33] compared the quality of experience of stereoscopic 3D videos in different projection devices, including flat screens, panoramic screens, and VR headsets. The study found that participants who used VR headsets experienced higher levels of CS than those who used flat or panoramic screens. Researchers [204] have also found that users who used a higher-end HMD experienced less CS than those who used a lower-end HMD. These findings suggest that careful consideration should be given to the selection of display type when designing VR systems, especially for applications where CS can have a significant impact on user experience and performance.

Type of input: Several input methods are used in VR, such as hand controllers, treadmills, and motion sensors, which can have different effects on CS. According to Mittelstaedt et al. [163], the type of input device used in a virtual bike simulator affected the level of CS experienced by the participants. The researchers found that the use of motion control, such as steering with handlebars, led to higher levels of CS compared to non-motion control, such as using a joystick. Monteiro et al. [164] conducted a study exploring the effect of 3D views and controller types on first-person shooter games in VR. They found that the use of a gamepad controller caused less CS than using a hand controller. The effect of the input method on CS is not limited to game-based scenarios. Overall, the method of input used to control movements in the virtual environment can have a significant effect on CS. However, more research is needed to fully understand the relationship between input methods and CS in different VR scenarios.

3.2.2 VR Content. The design of the virtual environment can have a significant impact on the likelihood and severity of CS. The virtual content-related factors are as follows:

Visual motion: The visual motion stimulus is considered to be one of the primary causes of CS [100], which is the movement experienced by the user in a virtual environment. Higher visual motion speeds have been found to increase the risk of CS [38, 119], and the way the content moves within the virtual environment can affect the level of CS experienced by users. One study conducted by Tran et al. [237] evaluated the impacts of multiple factors on four main quality of experience aspects of 360°videos, namely, perceptual quality, presence, CS, and acceptability. The results showed that the way the content moved within the 360°videos affected the level of CS experienced by the users, whereas the type of device used for rendering the video did not have a significant effect on CS. Another study by Keshavraj et al. [100] found the vection, or sensation of self-motion, to be the highest when peripheral stimuli contained motion, and they found a positive correlation between the vection and the visual motion stimulus. The severity of CS significantly

increases with higher height and faster speed in the virtual environment [38], but brief exposure to turbulent visual motion alone was not enough to induce CS [218]. Furthermore, a study [229] also found that CS increased with an increase in angular velocity, and it was more intense when rotating around X and Y axes compared to the Z axis.

Means of Locomotion: Studies have investigated the impact of locomotion methods in virtual environments on CS. A sensory mismatch can occur when the user's virtual movements do not match their physical movements, causing discomfort and disorientation. One aspect of locomotion that affects CS induction is the control over motion. A study by Venkatakrishnan et al. [241] found that control has a direct effect on presence and an indirect effect on CS through presence. A greater sense of presence in VR can lead to a lower likelihood of experiencing CS. However, another study [242] found that participants who had control over their motion experienced greater levels of CS than those who did not. They also found that hand-based motion control resulted in significantly less CS than whole-body motion control. This suggests that the type of control mechanism used in VR, as well as the extent of control, can also affect the likelihood of experiencing CS. Another aspect of locomotion is the type of transition. Teleportation (instant teleportation) was found to be the least sickness-inducing among the three transition techniques compared in a study by Rahimi et al. [191]. Several other studies [35, 121, 138, 191] have found teleportation to reduce the occurrence of CS compared to naturalistic free movement. Teleportation completely skips the visual transition period between two points, thereby reducing the sensory conflict. However, realistic walking-based locomotion, like natural walking, redirected walking, and walk-in-place, are preferred when physical exertion is not a factor [4].

Field of view: FOV has been identified as a factor that can contribute to CS in VR experiences. FOV refers to the extent to which the user's field of vision is captured by the display in a VR headset [157, 232]. A larger FOV results in a greater sense of immersion as more of the virtual world becomes perceptible to the user. This measurement is not fixed for HMDs, and users can adjust the distance between the eye and lens to achieve the desired FOV. Previous research has shown that reducing the FOV can be an effective way to alleviate symptoms of CS in VR [3, 13, 54, 140, 230, 258], as it reduces the disparity between the virtual and physical FOVs. However, excessively narrow FOV can lead to decreased immersion and increased head movements, which may exacerbate CS [22]. Therefore, finding an optimal FOV that strikes a balance between reducing CS and maintaining immersion is crucial for creating comfortable VR experiences.

Type of the content: Several studies have been conducted to investigate the relationship between CS and the type of environment presented to the user [8, 31, 66]. Action content, fast movements, and abrupt alterations in the VR environment were found to induce more CS symptoms than neutral and static content [67]. However, users with a high preference for adrenaline sports, such as extreme sports, are less likely to experience CS. Moreover, the genre of the virtual environment plays a significant role in CS, where the horror genre induces more sickness than the pleasant genre [31]. The perspective used in VR also has a significant impact on the likelihood of experiencing CS, where third-person perspective VR is generally found to be less likely to induce simulator sickness compared to first-person perspective VR [165]. Moreover, the geometry and terrain of the virtual environment can also affect CS. According to one study [8], users experienced a higher degree of CS when traveling in the presence of regular bumps and irregular terrain than when doing so on flat geometry. Additionally, changes in weather conditions significantly affect CS levels experienced by users in VR environments. Research using a helicopter simulator found that participants reported higher CS scores during stormy conditions compared to clear weather [2]. This increase in CS is attributed to the greater difficulty in maneuvering and increased motion intensity under adverse weather conditions, enhancing the sensory conflict that contributes to CS.

The knowledge from these studies can help VR developers to create more comfortable and user-friendly VR environments that minimize CS symptoms. However, further investigation is required to examine the specific mechanisms behind the relationship between CS and different types of virtual environment settings.

Duration: Several recent studies have highlighted the importance of exposure duration in predicting the severity of CS. For example, Risi and Palmisano [202] found that prolonged exposure to HMDs led to a greater incidence of CS. Similarly, Aldaba and Moussavi [5] reported that the severity of CS increased with VR exposure time when participants used different types of locomotive controllers during navigation tasks. Cao et al. [23] compared VR driving simulation using HMDs and desktop displays and found that longer exposure duration in the HMD condition led to higher levels of CS. Treleaven et al. [236] observed a similar trend during neck motion-controlled VR tasks and concluded that longer exposure duration increased the likelihood of incidence of CS. These findings suggest that exposure duration should be carefully considered when designing VR experiences to minimize the risk of CS. Additionally, Porcino et al. [185] used ML algorithms to identify the causes of CS in VR games and found that prolonged exposure time was one of the contributing factors. A study by Zhang et al. [270] reported that continuous exposure to live-action 180°videos resulted in higher levels of CS, increasing with exposure time. These studies demonstrate the importance of exposure duration in predicting CS and highlight the need to carefully monitor exposure time in VR experiences to reduce the risk of CS.

Visual complexity: One of the factors that contribute to CS is the visual complexity of the VR content. Visual complexity refers to the amount of visual information that the user is exposed to while using VR technology. Studies have shown that as the level of visual complexity increases, the likelihood of experiencing CS also increases. A study by Pouke et al. [188] reported that increased visual complexity was positively correlated with increased CS symptoms. These findings suggest that reducing the visual complexity of VR content can potentially reduce the incidence and severity of CS. This can be achieved by simplifying the visual elements of the VR environment or by optimizing the graphics settings to reduce the amount of visual information presented to the user.

4 Measurement of CS

With a clear understanding of CS causes, the next step is to examine how these phenomena are measured, detailing the different techniques that have emerged in recent research. The VR industry is still relatively new and growing. To improve the quality of user experience, it is essential to measure CS. Understanding the prevalence and severity of CS can help VR developers create more comfortable and enjoyable VR experiences, and also identify individuals who may be at risk for adverse reactions. Measuring CS can aid in research studies investigating the causes, effects, and potential interventions for CS. Furthermore, the use of VR in therapeutic contexts, such as managing pain, treating phobias, and addressing post-traumatic stress disorder, has been on the rise. Measuring CS can ensure the therapy is safe and effective for patients, and evaluate the effectiveness of countermeasures. Recent studies have focused a lot on measuring CS. After doing a thorough analysis of the available literature, we have identified the main components required for the measurement of CS (As shown in the Figure 3). We termed this whole process of measuring CS as **Cybersickness Evaluation System** (**CES**). The input to the CES can vary from a simple questionnaire to complex physiological signals like **heart rate** (**HR**) and brain activity. The CES then processes the input features and, based on the type of evaluation process used, produces an output measuring the state of the user with respect to CS. Based on the available literature, the various aspects of the measurement of CS are discussed as follows -

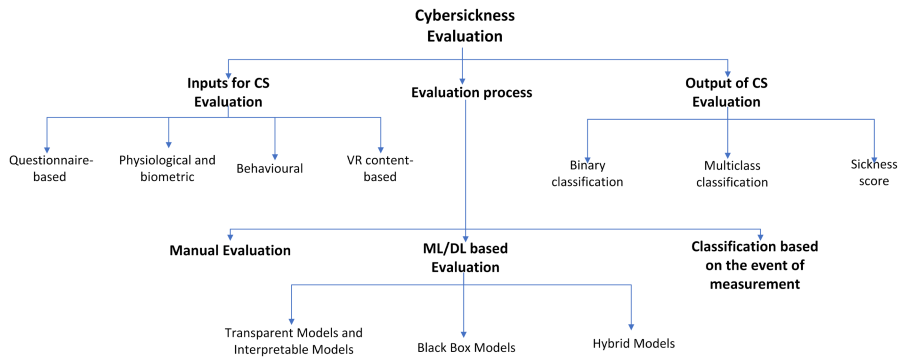

Fig. 3. Taxonomy of various measurement techniques of CS.

4.1 Inputs of Cybersickness Evaluation

The system's input quality determines the system's output quality. If the input data is incorrect, then we will never get the desired output. This concept is known as **garbage in, garbage out** (**GIGO**) in the computer science domain. Based on our survey, we classified the input for quantifying the sickness levels into questionnaire type, physiological type, behavioral type, and content type.

4.1.1 Questionnaire-Based Input. The subjective measurement of CS involves using self-report questionnaires or scales to ask users to rate their symptoms of discomfort or disorientation on a numerical scale. These questionnaires are the most commonly used tools for assessing CS because they are easy to administer and provide valuable information about a person's symptoms and experiences within virtual environments. This information helps researchers and practitioners better understand the causes and effects of CS and identify potential interventions. Questionnaires can also be administered to large groups of people, which allows for statistical analysis and generalization of results. Typically, questionnaires are administered before and after a VR experience, and sometimes even during the experience. The SSQ is the most extensively used questionnaire to assess the severity of CS in research studies. It consists of 16 items divided into three subscales—oculomotor, disorientation, and nausea. Participants rate the intensity of individual symptoms on a rating scale from 0 to 3, where a higher score denotes more severe symptoms. The scores obtained can be used to quantify the degree of CS experienced and compare the effects of different VR systems or conditions. There are several advantages as well as disadvantages of using SSQ to measure CS.

The advantages of using SSQ are as follows -

- Standardized measure: The SSQ is a well-established and widely used tool for measuring CS symptoms in VR environments.
- Multi-symptom assessment: The SSQ evaluates a variety of CS symptoms, such as oculomotor, disorientation, and nausea symptoms, allowing for a more comprehensive assessment of a person's CS experience.
- Easy to administer: The SSQ is a self-report questionnaire that can be administered before, during, or after exposure to a VR environment. This makes it easy to use in both research and clinical settings.
- Quantifiable results: The SSQ provides a score for each symptom, and an overall score, which allows for easy quantification of CS and comparison of results between different VR systems or conditions.

The following are the disadvantages of using SSQ -

- Self-reported: The SSQ relies on self-reported symptoms, which may be subject to bias or misinterpretation.
- Subjectivity: Symptoms are measured using a subjective rating scale that can vary from person to person.
- Not specific to VR: SSQ was originally developed to measure CS in flight simulators, so it may not be fully applicable to all VR systems.
- Limited symptom coverage: The SSQ does not cover all possible symptoms of CS, and it does not cover symptoms related to the cognitive or psychological effects of VR.

The SSQ was initially developed for military flight simulators to measure simulator sickness. It was designed using military aviators who were less susceptible to simulator sickness than the average population [222]. However, since it was not developed for VR, researchers have tried to modify the SSQ to measure CS in VR. Bouchard et al. [19] suggested changing the SSQ scoring guidelines to improve the accuracy of CS assessment in virtual environments. Kim et al. [104] developed a CS questionnaire (VRSQ), which consists of nine items, divided into oculomotor and disorientation categories. The VRSQ is quicker and more efficient than the SSQ. Other variations of the SSQ for measuring CS include the CSQ [227], which exhibited superior psychometric properties when used to evaluate the impact of VR applications for HMDs. Researchers have also developed shorter and more concise questionnaires to assess specific conditions, such as the **Misery Scale** (**MISC**) [18, 153, 240] and the **Fast Motion sickness Scale** (**FMS**) [96–98]. To measure one's susceptibility to CS, researchers have developed tools such as the **Motion History Questionnaire** (**MHQ**) [94, 123], the **Motion Sickness Susceptibility Questionnaire** (**MSSQ**) [62, 63], and the **Visually Induced Motion Sickness Susceptibility Questionnaire** (**VIMSSQ**) [99]. The Nausea Scale [215–217] is commonly used to assess overall discomfort, while forced-choice questions [32] and the **Motion Sickness Assessment Questionnaire** (**MSAQ**) [61] has also been utilized to do the same. Although the SSQ is the most widely used tool for the subjective measurement of CS, it should be used in conjunction with other measures to obtain a more comprehensive understanding of the effects of VR on the human body. Researchers have developed various other methods to measure and better assess CS symptoms.

4.1.2 Physiological and Biometric Input. Subjective measurement of CS heavily relies on communication from users, which may not always be reliable. However, advancements in technology and the availability of sensors have enabled the use of physiological and biometric inputs to detect CS. Objective measurements such as HR, **galvanic skin response** (**GSR**), and electroencephalography (EEG) have shown drastic changes during CS episodes [41, 82, 144, 198]. Researchers can use these signals, in conjunction with machine ML and DL models, to detect CS without relying solely on user communication. **Event-related potential** (**ERP**) components have been found to be affected by CS [1], indicating the possibility of analyzing ERPs in response to stimuli to indicate CS levels. EEG is widely used to detect CS, and studies have shown that certain electrodes can evaluate CS. One study found changes in brain patterns and less connection between some brain networks were observed after experiencing CS [93]. This study was the first to look at the brain's response to CS in such detail. Although they suggested that these brain patterns might be utilized to detect the incidence of CS, additional research is required to fully comprehend this. EEG is one of the most widely used physiological signals to detect the occurrence of CS [27, 69, 85, 106, 120, 132, 133, 135, 148, 167, 265]. DL methods have been employed along with EEG data to detect CS [87], with some studies analyzing EEG's power spectrum and raw waveform. HR and HRV measures obtained from **electrocardiogram** (**ECG**) have also been used to quantify CS [155], with several metrics proposed as potential indicators of CS. Other physiological signals

used to detect CS include **electrogastrogram** (**EGG**), **electrooculography** (**EOG**), respiration signals, **photoplethysmography** (**PPG**), **electromyogram** (**EMG**), GSR, and eye-tracking data. Researchers have derived multiple variables from these signals to quantify CS, including analysis of EEG's power band [42, 87, 101, 133, 135, 141, 159, 178, 198], electrical potentials of eye movement from EOG [41, 42, 59], HR, and its variability measures from ECG [41, 42, 59, 81, 82, 158, 172], gastric activity from EGG [41, 42], muscle activity from EMG [198], respiration rate [41, 42, 59, 81, 82, 172], and skin conductivity from GSR [41, 59, 81, 82, 158]. While there have been inconsistencies and contradictions in the information derived from these studies, the use of various physiological and biometric inputs provides promising avenues for detecting and quantifying CS. Further research in this area may lead to a better understanding of CS and the development of effective prevention and intervention strategies.

4.1.3 Behavioral Input. Postural sway has been widely used in evaluating CS [47, 153, 177, 223, 226]. Axial movements or changes in the user's **centre of pressure** (**COP**) during the VR exposure were monitored for postural sway. According to several studies, these metrics can either accurately predict the incidence of CS [30, 44, 244] or have a positive correlation with subjective self-reports [52]. Several studies came to the conclusion that there is a positive correlation between postural instability and severity of CS [40, 110, 146, 196, 233]. Several studies also focused on various eye-related factors to measure CS. As potential indicators, eyeblinks [41, 42, 59, 113], distance heterophoria [73], changes in eye position [43], vergence and accommodative responses [264] have all been studied. Jeong et al. [86] investigated the role of eye-tracking data in predicting CS. Chang et al. [29] developed a regression model to predict CS using only eye-related data. They found that natural gaze behaviors during a VR experience can serve as an important indicator for predicting CS. Wang et al. [245] used eye-tracking and character movement data for real-time detection of CS. Another study by Lopes et al. [151] explored the relationship between pupil position and blink rate with the severity of CS. Weight shifts [42, 247], head and waist movements captured from body sensors [42, 134] have also been explored as potential markers to measure CS. Generally, blink rate, head and body movement, and postural sway have all been found to be useful features for measuring CS.

4.1.4 VR Content-Based Input. The complexity of the visual environment in VR can contribute to CS. Researchers have tried to use different features related to VR content to evaluate CS. Padmanaban et al. [174] found that vection and CS are correlated as a function of relative motion depth. They also used FlowNet [46], a **convolutional neural network** (**CNN**)–based optical flow algorithm, and use its outputs to calculate their features for estimating CS. Optical flow and depth cues play a crucial role in the perception of self-motion and the establishment of a sense of presence in VR environments. Changes in these cues can lead to the experience of CS. Hence, they could be used as markers to identify CS episodes. Measuring the presence of these optical flow and depth cue conflicts can help researchers and developers identify the specific factors that contribute to CS, and work toward reducing its occurrence through design and technology solutions. Kim et al. [102] used exceptional motion of the VR video content to measure CS. They used deep convolutional autoencoder network for their experiment. Their findings suggested that one of the key causes of CS was exceptional motion velocity.

4.2 Evaluation Process

Upon receipt of the input, whether objective or subjective, it is imperative to process or interpret the data to assess CS. This evaluation process can be broadly categorized into two distinct methods—manual evaluation and ML-based evaluation. The former entails human intervention and cognitive processing, while the latter relies on automated algorithms and artificial intelligence.

The choice of evaluation method is typically dependent on the specific needs and context of the analysis.

4.2.1 Manual Evaluation. This category encompasses the evaluation of CS through the intervention of human cognition and processing. Typically, any assessment conducted by means of questionnaires, such as the SSQ, VRSQ, falls under this category [19, 71, 99, 104, 209, 212, 268].

4.2.2 ML Based Evaluation. The categorization of ML models based on their explainability has garnered significant attention in recent years. The classification framework includes four categories—Transparent Models, Interpretable Models, Black Box Models, and Hybrid Models. All types of models come with their own advantages and disadvantages. It is important to consider various aspects of the models while selecting the appropriate model for a particular application. We have collected various ML models employed to measure CS and we have presented them in (Table 3).

Transparent models and Interpretable models: Transparent models refer to those models that provide clear insight into the decision-making process, and the model's inner workings can be easily understood. However, the Interpretable models are similar to Transparent models, but they offer more flexibility in terms of customization and provide more precise details about the relationships between the input and output. As discussed in the previous section, researchers have found many objective inputs such as EEG, **motion-to-photon latency delay** (**MPD**), and postural stability, to predict the onset of CS. Many researchers [27, 65, 69, 85, 120, 132, 139, 253, 265] have used a transparent model-based approach to measure sickness from these inputs. The sickness was measured by observing the relationship with the CS. For instance, studies have shown that CS can lead to changes in EEG power spectra in certain frequency bands, such as theta (4-8 Hz) and alpha (8-13 Hz) bands [95]. In a study conducted by Dennison et al. [41], stepwise regression analysis was utilized to measure the occurrence of CS using a variety of physiological measures. Researchers have also used some interpretable ML algorithms, such as CN2 rule induction [178], Bagged Decision tree [42], Naive Bayes [159], KNN [134], LDA [42], LSTM regression analysis [81] to measure CS. In a recent study, Kundu et al. [116] employed a combination of ML techniques, including the **Explainable Boosting Machine** (**EBM**), decision tree, and logistic regression, to identify and provide insight into CS based on bio-physiological and subjective measurements obtained from users. Interpretable models help to build trust in the model and its predictions. It provides clear and understandable explanations of how they arrived at their predictions. Therefore, the use of interpretable ML models can improve transparency, trust, and accountability in ML systems.

Black Box Models: Black box models refer to models that cannot be easily understood or explained, and their decision-making process is shrouded in secrecy. Numerous techniques have been proposed in the literature to automatically detect CS using blackbox models based on ML/DL algorithms that leverage various objective and subjective data sources. An example of the successful employment of various algorithms in the realm of CS estimation is presented in the work of Jin et al. [107]. The researchers explored the utility of three widely used classifiers, namely **Long short-term memory** (**LSTM**), CNN, and **Support Vector Regression** (**SVM**), in estimating the level of discomfort in users. Their findings indicated that LSTM achieved the best results. Similarly, Lee et al. [127] presented a novel approach in their work [127] to enhance the accuracy of CS detection. Specifically, they introduced a 3D-CNN along with a multi-modal deep fusion method that integrated optical flow, disparity, and saliency features. Their approach yielded superior results in comparison to previous techniques. In another study, Jin et al. [106] used CNN and LSTM models to estimate the cognitive state based on brain signals and their relationship with CS levels. In contrast, Kundu et al. [117] used Kalman filtering techniques and LSTM to address the problem of CS.

Table 3. ML based Evaluation Methods of CS

Article	Algorithm	Input	Output
Kim et al. [106]	CNN, RNN, LSTM	EEG	Multiclass
Kim et al. [108]	CNN	VR Content	Sickness Score
Lee et al. [126]	ConvLSTM Regression	VR Content, EEG, ECG, GSR	Sickness Score
Islam et al. [82]	CNN-LSTM	ECG, Breathing rate, GSR	Multiclass
Kim et al. [107]	CNN, LSTM-RNN, SVR	Video features: Color, motion, Texture	Sickness Score
Li et al. [131]	ANN, Linear regression, SVM	VR Content, EEG, ECG, GSR	Multiclass
Jeong et al. [87]	DNN, CNN	EEG	Binary
Islam et al. [79]	LSTM, CNN	Eye tracking, Head tracking, Video	Sickness Score
Kim et al. [109]	LSTM	Video	Sickness Score
Kim et al. [109]	Stepwise regression, Random forest	EEG, ECG, EDA, Body, Skin, Respiration	Sickness Score
Weech et al. [250]	Principal component regression model	Sway path length, Balance control	Sickness Score
Kim et al. [111]	ConvLSTM	Video: Camera shake	Sickness Score
Jeong et al. [86]	CNN-Attention, BiLSTM-Attention	Eye tracking	Sickness Score
Du et al. [48]	3D-CNN	Video features: Saliency, optical flow, disparity	Sickness Score
Hell et al. [68]	RNN	Video vector	Sickness Score
Li et al. [134]	KN, RF, LR, MLPNN	EEG, Sway, Head and Body movement	Multiclass
Kim et al. [102]	Conv Autoencoder	Video vector	Sickness Score
Shang et al. [210]	DGCCA-SVM	HR, HRV, GSR< BR	Multiclass
Li et al. [129]	Linear Multiple Regression	EEG	Sickness score
Lee et al. [125]	ConvLSTM, DeConvLSTM	Content, EEG, EKG, GSR	Sickness Score
Chang et al. [29]	Linear Regression	Eye Tracking	Sickness Score
Chun et al. [34]	Random Forest	HR, GSR	Sickness Score
Anwar et al. [9]	LR, ANN	QPs, Resolution, Rendering device, Gender, user's interest, familiarity	Binary
Padmanaban et al. [174]	Bagged decision tree	Quantifying speed, direction, depth	Sickness Score
Islam et al. [80]	LSTM, MBEATS, DeepTCN	Eye and Head Tracking, HR, GSR	FMS Score
Kundu et al. [116]	EBM, DT, LR	Duration, Rotation, Acceleration, GSR, HR	FMS Score
Dennison et al. [41]	Stepwise regression	EEG, ECG, EOG, GSR, PPG Breathing and Blink rate	SSQ Score
Liao et al. [135]	SVM, MLP, CNN, LSTM	EEG	Binary
Porcino et al. [186]	RF, LMT	Duration, z-axis rotation, age, gender, experience	Binary and quarterly
Martin et al. [158]	SVM, GB, RF, LR	BVP, EDA	Binary, Multiclass
Kim et al. [105]	SVR	Head angular velocity, visual angular velocity, perceived angular velocity	Sickness Score
Li et al. [133]	KNN, Polynomial-SVM, RBF-SVM	EEG	Binary
Kuo et al. [118]	RFR	Hybrid Temporal feature	Sickness Score
Wang et al. [247]	LSTM	Postural sway	SSQ Score
Kim et al. [112]	CNN-RNN	Video features: Spatial, Spatio-temporal, rotational, exceptional motion	Sickness Score
Kim et al. [103]	CNN, ConvLSTM	Skin Conductance	Sickness Score

Despite their high accuracy, black box models require huge amount of data, processing power and they are difficult to interpret.

Hybrid Models: Hybrid models are a combination of different ML models, designed to provide high accuracy and transparency in their decision-making process. Lee and Alamaniotis [128] proposed a method that uses a **Deep Embedded Self-Organizing Map** (**DESOM**) and an Autoencoder designed from EEGNET, a specialized CNN for EEG prediction. They compared the DESOM method with KNN and SOM clustering methods and found that DESOM provided the most efficient index values with a purity of 0.96875 and an NMI of 0.42561. Another study by Li et al. [134] employed a voting classifier that used four base classifiers: Random Forest, KNN Classifier, Logistic Regression, and Multilayer Perceptron Neural Network for binary and three-level classification of CS. The voting classifier achieved an accuracy of 0.911 and kappa of 0.80 across 20 subjects. Binary classification of multiple subjects yielded an accuracy of 0.763, while three-level classification produced sensitivity values of 0.791, 0.504, and 0.867 and a kappa of 0.51. Hybrid models offer several advantages over the purely black box or transparent models, depending on specific application requirements. For example, hybrid models can potentially combine the strengths of both types of models by providing good predictive accuracy and some degree of interpretability. This can be useful in situations where accuracy, as well as interpretability, are essential, such as medical diagnosis. Moreover, hybrid models offer more flexibility in model design and optimization, enabling the use of different types of models and techniques best suited for different aspects of the problem at hand. For instance, a hybrid model can use a deep neural network for feature extraction and a decision tree for classification, allowing for more efficient use of resources and faster training. In conclusion, hybrid models provide a valuable tool in ML applications, allowing for the integration of various models' strengths and overcoming their weaknesses.

4.2.3 Classification Based on the Event of Measurement. The term "detection" refers to the process of identifying data associated with an ongoing CS event. On the contrary, "prediction" is defined as the act of forecasting future CS using data that precedes the event. Notably, most existing studies [41, 42, 101, 141, 159, 178] in the literature that employed physiological signals primarily focused on the detection of CS. Conversely, a few studies [41, 81, 82] utilized physiological signals that were focused on predicting CS. Researchers [11, 48, 80, 102, 103, 109, 111, 126, 139, 213] have also employed video features to predict CS. Limited research has been conducted on the ability to anticipate upcoming CS incidents by employing pre-VR immersion baseline measurements. Additionally, there is a scarcity of ML research concentrating on predicting the susceptibility or impending CS episodes by employing the typical physiological state of a person.

4.3 Output of CS Evaluation

The output of self-report questionnaires for CS is typically in the form of scores or ratings, which are based on users' subjective experiences of various symptoms such as dizziness, headache, nausea, and disorientation [19, 99, 104, 155, 209, 212, 268]. These scores are then used to assess the level of CS experienced by the user. In contrast, the output of ML-based CS detection and prediction models is typically in the form of binary or multiclass classification labels or a sickness score, which is calculated using features extracted from physiological signals or other data sources. The classification labels or sickness scores indicate the likelihood or severity of CS being experienced by the user based on the input data provided to the model. For instance, the study by Islam et al. [79] classified the sickness into Low Sickness, Moderate Sickness, and Acute Sickness. However, in the study by Khoirunnisaa et al. [101], the sickness was classified as either sickness or no sickness. On the other hand, Dennison et al. [41] outputted the sickness score using a regression model, which quantifies the severity of CS experienced by the user.

5 Work on CS Mitigation

Equipped with insights into measurement approaches, this section discusses effective strategies for mitigating CS, highlighting interventions from both developers' and end-users' perspectives. CS can wreak havoc in the VR space if left unattended. If we are unable to completely prevent CS from occurring, we can at least strive to lessen its impact. It is essential to mitigate CS to improve user experience, protect user health and safety and also increase the adoption rate of VR technology to ensure the continued growth and success of the VR industry. Proper measures must be implemented at both the user and developer end to lessen the impact of CS. Studies have shown that CS mitigation strategies often come at the cost of presence in the virtual environment. This is where the challenge lies while developing CS mitigation strategies without compromising on the user experience in the virtual world. In this article, we have tried to put forward various mitigation strategies found across the literature under one roof, both from the perspective of developers as well as the end users.

5.1 Mitigation Strategies from Developers' Perspective

From a developer's perspective, various strategies have been conceived in an attempt to mitigate CS. Proposed solutions range from simple virtual environment modifications to complex hardware alterations. Due to the subjectivity associated with CS, it is highly unlikely that a single approach will be able to reduce the impact of CS on all the participants. Here, we'll talk about a variety of intervention strategies that developers might use to address the issue of CS.

Adjusting the FOV is a widely studied technique to reduce CS. FOV describes the amount of the virtual environment visible at any moment. Altering the FOV has proven effective in mitigating the impact of CS [3, 13, 54, 140, 230, 258]. In VR, restricting the FOV, known as "vignetting" or "tunnelling," can alleviate CS symptoms by reducing peripheral optical flow, which is believed to exacerbate CS symptoms [52, 114, 131]. Such restrictors are effective across genders without negative impacts, as noted by Zayer et al. [3]. However, these strategies can reduce the sense of presence and impair performance in visual tasks [208, 243]. The study by Teixeira and Palmisano [230] serves as a case study, demonstrating that dynamic FOV restrictions in a commercially available HMD game significantly reduced CS severity compared to unrestricted FOV conditions. Their findings support the use of dynamic FOV adjustments to mitigate CS, highlighting a balance between reducing CS symptoms and maintaining immersion and presence. This balance is crucial as both too small and too large FOV sizes can, respectively, break immersion or induce sickness. Despite the effectiveness of dynamic FOV in reducing CS, developers face challenges in optimizing FOV size to prevent immersion loss without inducing CS. Furthermore, although dynamic FOV restriction has been shown to be an effective countermeasure for CS, its implementation must be carefully considered to avoid diminishing the user's experience. The potential for reduced sense of presence and decreased visual search performance underscores the importance of a balanced approach in applying FOV restriction techniques. Developers are encouraged to consider these aspects to ensure that mitigation strategies do not inadvertently compromise the overall VR experience. Generally, a symmetric mask that occludes the periphery in reaction to translational and/or angular velocity is used to implement the classical FOV restrictor. Wu and Rosenberg [259], however, used an adaptive restrictor that utilized an asymmetric mask to obscure regions of the periphery with the higher optical flow during virtual locomotion while leaving regions with lower optical flow visible. Their approach was effective in reducing CS and increasing immersion time in the virtual environment. When compared to symmetric restriction, presence ratings were much higher when employing the adaptive restrictor. A novel variant of the FOV restrictor termed the *side restrictor*, was proposed by Wu and Rosenberg [258]. This approach utilizes an asymmetric mask to obscure

only one side of the periphery during virtual rotation and laterally shifts the center of restriction in the direction of the turn. Results suggest their approach to be effective in mitigating CS during frequent virtual turns while showing no drawback when compared to the symmetric restrictor. Wu et al. [256] also explored the idea of using an asymmetric FOV restrictor called *ground-visible restrictor* to mitigate CS. They ensured that the ground plane remained visible during movement and found that this approach provided users with a comfortable experience, postural stability, and a subjective sense of presence without any apparent negative effects associated with maintaining the visibility of the ground plane during FOV restriction.

Hillaire et al. [70] researched how users moved their eyes while being in a virtual environment and suggested a model of dynamical visual blur based on two different types of blur effects, namely **depth of field** (**DOF**) blur and a peripheral blur. Gamers' performance was not affected by this technique and it managed to enhance their enjoyment of the game. Nie et al. [170] proposed an approach to reduce CS by applying dynamic blurring on a user's retina. Their research aimed to create a non-pitch-black display in view, which is different from the conventional approach. The experimental results are consistent with the idea that CS can be minimized by dynamic non-salient region blurring. Lin et al. [145] studied how the presence and size of a static central window in peripheral FOV blurring affect CS. They used a Gaussian blur effect in the display peripheral region, provisioning a full-resolution central window. Their observations suggested that the use of peripheral FOV blurring was effective in reducing CS.

DOF blur effects are well-known depth cues in human vision. One of the factors that contribute to CS is the mismatch between the user's visual perception and the body's vestibular system, which is responsible for maintaining balance and spatial orientation. DOF blur can help to reduce this mismatch by mimicking the way that the human eye naturally focuses on objects at different distances, which can help to create a more natural and immersive experience. The blur effect can help to smooth out these visual cues and make the VR experience more comfortable and less disorienting for the user. Researchers have tried to manipulate DOF to mitigate the adverse effects of CS. In relation to CS, studies have shown that there is a difference in the processing of DOF between humans and the hardware, e.g., HMD [49, 72]. This disparity leads to eye-related symptoms, causing a strain on eye movements [25, 49, 72]. Carnegie and Rhee [25] studied the effectiveness of dynamic DOF blur in reducing the impact of CS. Participants in the study reported a decrease in the severity of symptoms brought on by HMD use, proving dynamic DOF's efficacy in alleviating uncomfortable symptoms of CS. Hussain et al. [75] developed a technique to incorporate spatial blur in VR systems to reduce CS. The technique makes use of concepts from foveated imaging and DOF. The result of the study showed that their foveated DOF effect was effective in reducing CS in virtual settings by lowering the sickness scores by almost 66%.

Another widely used technique to reduce CS is the usage of "rest frames". Rest frames refer to the fixed elements within the virtual environment that remain stationary relative to the real world, regardless of the user's movements. Rest frames that are attached to the head reference frame are known as independent visual backgrounds. A few examples of independent visual backgrounds are as follows—clouds [143], grids [50, 51, 169], and virtual nose [252, 255]. Studies have shown these strategies to be effective in reducing CS, although some contradicting studies also exist [271]. Few other varying implementations of rest frames have also been proposed, such as a wireframe model of the physical world in the periphery [257], a small reticle superimposed over the centre of the visual field [15] and an independent background that fills the entire FOV [50]. The use of dynamic rest frames to mitigate the effects of CS was investigated by Cao et al. [24]. In their study, dynamic rest frames were employed, which altered the opacity of the rest frame based on visually perceived motion as users traversed the virtual environment. Wu and Rosenberg [257] devised a method to combine dynamic FOV altercation with rest frames generated from 3D scans of the

physical environment. Although their study lacked the required breadth of testing, the initial result suggested their approach to be promising in reducing CS as well as improving the safety of the user. Lin et al. [142] used a Virtual Guiding Avatar to alleviate the symptoms of CS, which offered visual indicators of the impending virtual motion and a rather stable cue.

Previous research showed the possibility of reducing CS by manipulating optical flow [57, 83, 84]. Park et al. [180] proposed an interesting approach to reduce CS by visually mixing the optical flow patterns that are in the reverse direction of the virtual visual motion. The experimental outcomes showed that the suggested approach reduced CS in a statistically meaningful way. Buhler et al. [21] also investigated the effect on CS by reducing optical flow in peripheral vision. Bala et al. [12] also discussed the role of optical flow in the reduction of CS in 360°videos. They found that, despite being intrusive in terms of presence, a restricted FOV aligned with the optical flow is desirable for lowering CS and providing a safe VR experience. Lou et al. [152] tried to reduce the perceived optical flow by means of geometrical simplification of the virtual environment. A pilot study showed the potential of their method in reducing CS.

Hardware alterations have been one of the well-known strategies employed to combat the problem of CS. These hardware interventions can range from simple modifications to already existing VR hardware to coming up with new pieces of hardware altogether. The sparse peripheral display idea was introduced by Xiao and Benko [260]. It entails expanding the high-resolution HMD's existing periphery with a matrix of colorful LEDs. The outcomes revealed that this strategy reduced CS effectively. Andre and Coutellier [7] also tried to reduce CS with the use of miniature LED screens placed in the HMD. Ikei et al. [76] proposed a novel camera setup named TwinCam to mitigate CS. Experimental results suggest their mechanical camera setup can reduce motion blur and CS. Ra et al. [190] also proposed an idea of a novel hybrid 360°panoramic video camera rig. One of the causes of CS in the application to photon latency also called MPD, i.e., the time duration from the user's inputs to the displayed images. Achieving the ideal MPD would be a huge step in mitigating CS. But hardware limitations have made it incredibly challenging to solve the issue. Xie et al. [261] proposed a Processing-In-Memory based **asynchronous time warp** (**ATW**) design to achieve the ideal MPD. The implementation of their solution could be based on building on-chip ATW units, or it could also be based on software techniques. As technology is advancing at a rapid pace, more research is required to be conducted to overcome the limitations of the current generation's hardware.

Galvanic Vestibular Stimulation (**GVS**) has been shown to be effective in reducing CS and creating a more enriching user experience [64]. In VR, GVS is utilized to simulate missing vestibular cues or to simulate vestibular organ vibrations by adding noise to the sensory modality. Sra et al. [219] developed a tiny, lightweight wearable apparatus that uses GVS to lessen the impact of CS. A study by Weech et al. [251] showed that the severity of CS was lessened when noise was injected into the vestibular system using noisy-current GVS during VR immersion. Weech et al. [249] proposed reducing CS with the help of noisy vestibular stimulation through bone vibration.

Liu et al. [150] proposed a new approach called *PhantomLegs* that applies alternating haptic cues that are synchronized to users' footsteps in VR with the help of additional hardware. Findings indicate that the method significantly lowers CS during the initial exposure without compromising the comfort of the user. Jung et al. [89] proposed that floor vibrations that try to match with the vibrations of the virtual environment will reduce the conflict between visual and vestibular systems which, in turn, would also reduce CS. Their approach was based on introducing vibration-driven whole-body tactile feedback. Sawada et al. [206] explored the idea of reducing CS by simultaneously presenting sound as well as vibration in virtual driving scenes. Peng et al. [183] used two small vibration motors behind the ears to give unobtrusive vibrotactile feedback for VR walking experiences which managed to provide a more realistic experience as well as reduction of CS. Their

approach was called *WalkingVibe*. Yoon et al. [267] suggested using a motion platform to reduce CS. Results showed a reduction in CS by using the motion platform to synchronize the input transmitted through the sensory organ with the visual pictures transmitted through the HMD. A study by Rietzler et al. [201] showed that short bursts of rotational acceleration to match the visual forward acceleration were effective in reducing CS.

The locomotion technique is very critical for a good VR experience. There are various kinds of locomotion techniques available in the literature. Cmentowski et al. [36] proposed a novel walking method to reduce CS. Teleportation was somewhat preferable to steering locomotion as a user-controlled navigation strategy since it can lessen the symptoms of MS, as demonstrated by Farmani and Teather [53]. Freitag et al. [56] employed dynamic speed in steering to "skip" certain locations when going over long distances to avoid making the experience uncomfortable. Farmani and Teather [53] minimized the optic flow and corresponding visual-vestibular conflict by rendering the moving picture in discrete chunks rather than the individual frames that would typically be presented during a rotation. They called their approach *viewpoint snapping*. Dorado et al. [45] found that moving up and down the stairs was a problematic task in the virtual environment. In order to diminish users' perception of movement when traveling through stairs and as a result, the symptoms of CS, they suggested using invisible ramps on top of the stairs.

Among other miscellaneous approaches, Hu et al. [74] devised a novel real-time, online optimization for virtual camera control to reduce CS. Isaza et al. [77] proposed a novel rendering technique utilizing two different cameras to create a hybrid rendering that includes monoscopic and stereoscopic systems, called ***Dynamic Mono-Stereoscopic Rendering System*** (**DMSRS**). According to the results, DMSRS was able to reduce CS without compromising the user experience. Developers have tried to tackle the issue of CS from varied perspectives, with some being more promising than others.

5.2 Mitigation Strategies from Users' Perspective

In order to alleviate the negative symptoms of CS, users also need to take some countermeasures to deal with the issue. Subjectivity involved in CS makes it a very complicated problem to deal with. Different people have varied degrees of susceptibility toward CS. Hence, interventions only from the developer's end might not be enough. It is well established that increased exposure time in the virtual environment leads to severe symptoms of CS [221]. Hence, it is suggested to avoid continuous exposure for longer periods of time in the virtual environment. Through their experimentations, Kourtesis et al. [115] suggested that the maximum duration of a VR session should be between 55 and 70 minutes. Another observation is that the intensity of uncomfortable symptoms of CS gradually decreased in users after repeated experiences in the virtual environment [55]. The success of habituation depends on repeated short-duration exposure until undesirable symptoms appear, followed by repeated longer-duration exposure after full recovery. Unlike other potential approaches, habituation is a long-term process. But some users who experience the unpleasant symptoms of CS stop using VR technologies altogether without making any further attempts at adaptation. That could lead to a very poor adoption rate of VR in the long run. Hence, tackling the problem of CS is of utmost priority in the VR industry.

Stromberg et al. [228] explored the effectiveness of slow **diaphragmatic breathing** (**DB**) in reducing CS. The findings point to the potential advantages of slow DB approaches in mitigating CS in VR. Another study by Russell et al. [203] showed that DB enhanced parasympathetic nervous system tone, lowered respiratory rate, and minimized the onset of CS symptoms. Controlled breathing is consistent with the idea of triggering parasympathetic activity, which is thought to be consistent with a decrease in stress or anxiety during mildly nauseating exposure. Another interesting approach to reducing the uncomfortable symptoms of CS is by using acustimulation

[149]. This approach being a nonintrusive as well as an economically friendly solution could be a potential approach to mitigating CS in the near future, although more research is required.

One of the most accessible, easy-to-use, and effective ways to reduce CS is through the use of over-the-counter medications, provided that they are started long before the anticipated exposure. In certain circumstances, users may use anti-nausea and anti-emetic drugs to relieve their symptoms. These drugs have been effective in combating MS over the years. When adequate time to acclimatize to the virtual environment cannot be allowed, medical intervention may be advised. Iskenderova et al. [78] found surprising results when exploring the effects of alcohol on CS. They found that Alcohol consumption at a blood alcohol concentration of about 0.07% considerably lessened CS symptoms. Although an interesting find, more study is required for it to be considered a safe option to alleviate CS symptoms.

Users can also turn their attention toward some easy home remedies to reduce their symptoms. Kaufeld et al. [92] studied the effect of chewing gum on CS and found that chewing gum was able to reduce CS. Ginger root is another quick home remedy that may be used to relieve the symptoms of CS. Ginger root is a cost-effective, easy-to-find, side-effect-free solution that has shown a potential to alleviate symptoms of CS [137], although contradicting studies also exist [207]. Ranasinghe et al. [192] explored the possibility of using olfactory stimuli to reduce CS. Results showed that peppermint aroma was able to reduce CS in the participants. Although peppermint fragrance can lessen CS, the lack of any appreciable improvements in nausea-related symptoms described also indicated that extended exposure to olfactory stimuli may be a factor in the development of nausea and/or CS symptoms. Keshavarz and Hecht [98] explored the idea of reducing CS by introducing music. They found that only music perceived as pleasant irrespective of the genre of the music was able to reduce CS. This is an effective, low-cost, and easy-to-use method that should be explored further. An experiment by Paroz and Potter [181] suggested that external airflow was able to reduce CS as compared to control and reduce room temperature. They hypothesize that because the airflow cannot be used to counteract the virtual stimuli, it may have a tactile "white noise" effect, lessening the impact of sensory conflict. This airflow can be produced without the need for specialized hardware or software integration, which may provide VR users with a simple way to lessen the symptoms of CS.

A study by Mao et al. [156] found that in contrast to negative framing and broad cautions, positive framing not only lessens nocebo side effects but also completely eliminates them. Hence, using positive attribute framing to minimize nocebo side effects may be both affordable and moral. The authors proposed more studies to investigate the processes underlying these effects and evaluate if the advantages of positive framing extend to general and clinical groups. Smyth et al. [214] suggested undergoing specific training to improve visuospatial skills in order to reduce the effects of CS.

6 Discussion

CS has been a hot topic for discussion in the VR community. It has the potential to disrupt a multibillion-dollar industry. The adoption rate of VR heavily depends on how well we can tackle the problem of CS. Numerous studies have been conducted to develop a better understanding of the problem. In this article, we have done a thorough analysis of the three most critical aspects of CS—causes, measurement techniques, and mitigation strategies of CS. Internal factors, such as human factors like age, gender, prior experience with VR technologies, physical conditions, and emotional conditions, and external factors like hardware and VR content-related aspects, can induce CS symptoms. The characteristics of moving images, such as velocity, rotational pattern, and oscillation amplitude, have also been studied to understand their relationship with inducing VR sickness. Self-reporting-based questionnaires and physiological, behavioral, and content-based inputs are some of the techniques used to evaluate CS. Objective measurement is crucial to understanding

the issue of CS since not everyone experiences CS the same way. The process of evaluation can be manual, based on self-reporting, or objective, based on the machine-based evaluation. The mitigation strategy for CS can be examined from two perspectives, that of the developer and that of the user. Developers may utilize a variety of techniques such as FOV alterations, DOF blurring, introducing rest frames, manipulating of optical flow, and some hardware alterations to minimize the occurrence of CS. On the other hand, users may adopt certain practices to reduce the effects of CS such as limiting exposure duration, taking medication as needed, practicing DB, and employing simple remedies such as listening to music, consuming ginger root, and chewing gum. Researchers must continue studying the causes and symptoms of CS to design better solutions to mitigate this problem. Based on our analysis of the available literature, we have also provided a set of mitigation guidelines for both users and developers (Section 6.2).

6.1 Research Gaps

Despite being a well-known issue in the VR community, there are still gaps in the research on CS. After critically analyzing a carefully curated list of 223 highly relevant research papers on various aspects of CS, we have identified several research gaps in the field (Table 4). Addressing these research gaps will enable us to advance our understanding of CS and develop effective countermeasures for it.

6.2 Guidelines for Mitigating CS

Most Immersive VR experiences come with some sort of undesired symptoms of CS. In a study by Regan and Price, [199], more than 60% of the 146 participants reported symptoms of CS at some point throughout a 20-minute immersion and 10-minute post-immersion phase. This statistic suggests that it's highly likely that a user will experience some sort of undesired symptoms of CS during longer exposure in the virtual environment. Researchers have still not found a way to stop CS completely. In fact, as of now, we are not even close to completely uprooting the problem of CS. Hence, it's extremely important to reduce the impact of CS as much as possible. Otherwise, users will not adopt VR technology, owing to the terrible user experience due to CS. Hence, there is a need for preventive measures to tackle the problem of CS. It's crucial for the developers to adhere to a few rules when producing VR content, as well as for the users when consuming that VR content. After a thorough review of the literature, we are proposing a set of mitigation guidelines for both the developers as well as the users. These set of guidelines, when followed, will be able to reduce the impact of CS, as backed by several studies found across the literature.

6.2.1 Developer Guidelines.

(1) Developers should allow for adjustable IPD in HMDs as studies show smaller IPD in females lead to poor HMD fit and increased susceptibility to CS [220].
(2) Reduce application to photon latency, also known as MPD [225].
(3) Repetitive patterns (like gratings) and high spatial frequency content (like stripes, tiny textures, etc.) should be minimized or avoided as they might cause discomfort and feelings of vection. Flatter textures should be preferred to patterned ones, such as those that are solid colors [184].
(4) Limit or eliminate the usage of elements that create vertical acceleration (such as stairs), as horizontal optical flow when climbing (vection) and vertical acceleration can both cause symptoms. Ramps might be a better alternative, but they should only be utilised occasionally [57].
(5) To lessen the symptoms of CS, rest-frame graphical cues can be displayed in the foreground as well as panorama elements in the background [24, 179].

Table 4. Research Gaps

Research topic	Title	Remarks
Causes of CS	Role of the physical environment in Inducing CS	There is a lack of research on how physical environment elements like room size, temperature, ventilation, and lighting plays a role in inducing CS. Knowing how these elements play a role in inducing CS will enable us to come up with strategies to mitigate CS by altering the physical environment elements accordingly.
	Role of ethnicity in causing CS	Although some studies [248] have looked into how ethnicity affects CS, there is still a requirement for more study. Ethnicity may be able to explain some degree of subjectivity associated with CS.
	Role of gender in inducing CS	Although some studies [166, 168, 220] found that gender plays a role in inducing CS, there have been some contradictory studies suggesting that gender does not play a crucial role in causing CS. Further research is required on this front. Also, to remove any gender biases in the study, there is a need to conduct gender-neutral studies.
	Role of visual disability in inducing CS	It is reasonable to assume that users of VR technology with impaired binocular function caused by issues with convergence would experience greater oculomotor side effects than those with normal vision. Further research is required to establish vision impairment as a potential factor in causing CS.
	Role of the genre of VR content in inducing CS	There is a lack of uniformity in genre classification systems and the need for more qualitative research. More research is needed to understand the relationship between the genre of the content and CS.
Measurement of CS	Lack of standardized metric	There is a lack of standardized metrics present in the literature that can objectively compare the effectiveness of different measurement techniques. The lack of a standardized metric makes it difficult to compare different studies.
	Unavailability of large, quality datasets	Lack of data in large quantity is a bottleneck in measuring CS accurately. Without acquiring data in larger quantities, we would not be able to leverage cutting-edge DL methods to measure CS. Also, a lack of data will be a roadblock in developing a model that is able to generalize well.
	Lack of explainability of ML/DL models	Although Kundu et al. [116] proposed predicting and detecting CS using explainable ML (xML) models, there is a lack of research regarding using xML to measure CS. The trustworthiness of ML/DL models can be greatly increased by having inherent interpretability, which can give an understanding as to why a certain conclusion was reached by the model. This will also help researchers understand which factors contributed more towards CS, and accordingly, mitigation strategies can be proposed.
Mitigation of CS	Lack of predictive models for CS	Majority of the study we reviewed used models to detect CS from physiological signals. We observed that there is a scarcity of studies in the literature concerning the prediction of CS using physiological signals. Predicting CS symptoms before it actually happens would allow both the developers as well as the users to take adequate preventive measures.
	Investigation of various eye-tracking data as features to measure CS	As there is a number of eye-tracking features available, e.g., pupil diameter, pupil position, gaze direction, gaze origin, eye openness, etc., there is a need to study how different combination of eye features can be used to measure CS. Further studies are required to understand the significance of each component of the eye-tracking data.
	Using only sensor data of commercially available HMD to measure CS	Research by Islam et al. [79] suggested that it is possible to measure CS using only the data available from sensors present in commercially available HMDs. Further research is required on this front. One suggestion for the future is to integrate controller data along with HMD sensor data to measure CS. Further research can also be done on the real-time prediction of CS by integrating VR video content data along with sensor data.
	Use of incremental (online) learning	ML models would greatly benefit if we incorporate incremental (online) learning, making the models more robust.
	Lack of CS-specific medication	Over-the-counter medications often prescribed for CS, are primarily for MS. Since there is a significant difference between MS and CS, it is suggested to develop CS-specific medications. To test the efficacy of novel medications and their delivery methods, further investigations are required.
	Controlled breathing as mitigation technique of CS	Controlled breathing has been proven to be helpful in reducing CS symptoms [203, 228]. More research is required to understand how controlled breathing helps in the reduction of CS and further optimize the technique to increase its effectiveness.
	Role of soothing environment elements	More research is required on the usage of music and pleasant aroma as mitigation techniques of CS.

(6) Implied acceleration and deceleration in the optic flow should be kept to a minimum [152].
(7) As hyper-realistic graphic is known to induce CS, developers should provide the functionality of dynamically downgrading the rendering quality of the VR content in order to mitigate CS [246].
(8) Depth cues that try to mimic real-world cues can be used to mitigate CS. For user tasks, blurring less significant areas of the image may also be helpful in reducing the impact of CS [49, 70].
(9) Maintain a stable frame rate for smooth movement in VR, which can reduce the risk of CS [255].
(10) Provide comfort settings in the VR application to give users the ability to adjust the intensity of certain effects, such as motion blur [25], and subsequently minimize the risk of CS.
(11) Inform the user if the VR content will invoke CS symptoms or not (e.g., using CS prediction methods like [174] and [80]). This way, users can make a choice to participate and adequately prepare themselves for the VR experience.
(12) Notify the VR user about their exposure time in the virtual environment and accordingly give them suggestions about avoiding CS symptoms [115].
(13) Design shorter duration VR content so that users can enjoy them in one sitting and reduce the risk of developing severe CS symptoms [115].

6.2.2 User Guidelines.

(1) User should take short breaks in between VR sessions. A single VR session should not be very long (typically 15–20 minutes) [115, 202].
(2) New users of VR technology are advised to start slowly and gradually build up their exposure. They should begin with shorter sessions and gradually increase the time spent in VR [202].
(3) While trying out VR technology for the first few times, it is advised to have someone by their side so that in case of emergency, they can take the necessary steps.
(4) Find a comfortable HMD that fits the user properly and maintains proper hygiene of the HMD [220].
(5) Avoid sudden, rapid head movements while using VR as that can cause discomfort and contribute to CS [111].
(6) Users should avoid indulging in VR technology if they are carrying some physical illness [6, 110].
(7) Users should take preventive medicines, like anti-nausea and anti-vertigo medicines, when they start to feel symptoms of CS. These medicines take time to work. Hence, preventive measures should be taken as soon as the user feels uncomfortable in the virtual environment [269].
(8) Certain VR content may induce CS symptoms in only a specific group of VR users. As CS is extremely subjective, it's important to understand your own preferences and accordingly decide whether to avoid or indulge in certain VR content [231].

Building upon the insights provided in this survey, researchers can further advance the field of CS by delving deeper into the identified causes, measurement techniques, and mitigation strategies. This approach enables them to uncover additional gaps, including those highlighted in the survey, and conduct targeted research to develop more comprehensive solutions. By addressing CS in a holistic manner and refining existing mitigation guidelines, researchers can significantly enhance the overall user experience in VR environments, potentially increasing the technology's adoption and applicability across various sectors.

7 Conclusion

This comprehensive review has methodically addressed several pivotal research questions concerning CS in VR environments. Through the analysis of 223 studies, significant outcomes have been distilled regarding the causes, measurement, and mitigation of CS, alongside the identification of critical research gaps and future directions. Addressing the first research question (RQ1), the investigation identified a range of internal and external factors as leading causes of CS in VR environments (RQ1.a). Factors such as age, gender, prior VR exposure, hardware specifications, and VR content characteristics significantly contribute to the occurrence of CS. The impact of these causes varies across different VR applications (RQ1.b), influencing the severity and type of symptoms experienced by users, thus affecting user experience and the adoption rate of VR technologies. Concerning the measurement of CS (RQ2), the study highlights the evolving nature of VR technologies and the necessity for measurement techniques that can adapt to these changes. Current methods range from subjective questionnaires to objective physiological and behavioral assessments. However, the review underscores the lack of standardized metrics, which complicates the consistent evaluation of CS across different studies and VR platforms. In terms of mitigation strategies (RQ3.a), various developer and user-focused interventions have been explored. Developers have options such as adjusting FOV, altering graphical details, and modifying hardware configurations to minimize CS. Users can adopt practices like managing exposure duration and using preventive medications. The applicability of these measurement techniques across the diverse spectrum of VR applications (RQ3.b) remains challenging, suggesting a need for flexible and versatile solutions that can be tailored to specific user interactions and VR environments. Lastly, the article sheds light on emerging research gaps (RQ4.a) and their implications for future technological developments (RQ4.b). While significant strides have been made in understanding and managing CS, gaps remain in areas such as the influence of environmental and individual differences on CS susceptibility and the development of predictive models using advanced ML and DL techniques. Addressing these gaps could lead to more refined and proactive approaches to mitigate CS, enhancing the VR experience and expanding its application in various fields.

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